

Ignacio Alarcon Varela

Dublin - A94YW89 - Bolivian - Proficient English (EILST Certified)

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EDUCATION

College of Computing Technology Dublin - Bachelor of Science in Computing

2021 – 2025

Overall Result: 2:1

Technological University of Dublin (TUD) - Master of Science in Computing (Data Science)

2025 – 2027

Overall Result: On going

TECHNICAL SKILLS

Languages

Proficient: Python, Java, PostgreSQL

Experience: MongoDB, MySQL, JavaScript, R

Frameworks & Libraries: Spring Boot, Flutter, TensorFlow, Flask, PyTorch, React, Keras

Tech Stack: Ubuntu Server, RHEL, AWS, Git, Docker, Proxmox

PROJECTS

[Deep Learning Model for Fake Image Detection](#) | *Python, TensorFlow, Keras, Streamlit* | [Live App](#)

(Final Year Project — 3rd Place, [HECA Student Research Awards 2025](#))

- Developed model using Python, TensorFlow, and Keras to detect manipulated images and combat the spread of disinformation.
- Evaluated and compared various CNN architectures, including ResNet50 and EfficientNetB0, to optimize for the highest performance.
- Achieved a final model accuracy of 86.5% and deployed it as an interactive web application using Streamlit.

[AI Integrated E-Commerce Backend](#) | *Java, Spring Boot, Spring AI, Gemini, PostgreSQL, Docker, AWS* | [Live App](#)

- Architected a strong **Spring Boot** backend with a normalized **PostgreSQL** schema to handle secure product management and complex transactional workflows.
- Integrated Gemini as an auxiliary tool to accelerate content creation, allowing for the automated generation of product descriptions.
- Developed a React dashboard to serve as a client for the REST API, validating end to end data transmission and service logic.
- Containerized the platform using Docker and deployed it to AWS, establishing a scalable, cloud-native foundation for the application.

[Big Data Processing & Web App](#) | *Python, Java, Spring Boot, Hadoop, HBase, MySQL, Linux*

- Architected a scalable ETL pipeline using Hadoop MapReduce and Python to process and clean a 3.5GB dataset (25M+ records), migrating unstructured data into HBase and structured data into MySQL.
- Engineered a secure MVC web application using Spring Boot and Spring Security to visualize the data, implementing server-side pagination and dynamic search to optimize query performance for large datasets.
- Conducted rigorous database performance testing using YCSB (Yahoo! Cloud Serving Benchmark), analysing read/write latencies across varying workloads (A, B, and C) to tune the storage architecture.

[Serverless URL Shortener](#) | *Python, Flask, Vercel, Supabase* | [Live App](#)

- Developed a serverless microservice using Python and Flask, deployed on Vercel to keep infrastructure costs at zero.
- Integrated Supabase to create a reliable storage layer capable of handling read/write operations for thousands of links.
- Created stateless RESTful endpoints to manage URL generation and redirection, ensuring the app is always available with fast response times.

AWS cloud practitioner: [Certification Badge](#)

WORK EXPERIENCE

Chef de Partie, The Merrion Inn, Dublin

Nov 2022 – Present

Gastropub awarded Best Food Pub at the Irish Pub Awards 2023

- Collaborated closely with a team of 5 chefs to efficiently divide kitchen tasks and ensure smooth food prep before and during busy service hours
- Adapted quickly under pressure, handling last-minute order changes and special dietary requests, including allergies and celiac needs

Kitchen Porter, The Merrion Inn, Dublin

Sep 2021 – Nov 2022

- Handled kitchen deliveries, checked inventory and maintained high standards of cleanliness by washing and organising cutlery, dishes, and kitchen wear

Carer Assistant, Dublin

Dec 2020 – May 2022

Private Home Care Assistance

- Provided exclusive care by assisting one single client with daily living activities, supporting emotional well-being, monitoring health conditions and reporting back to medical staff and family members

References available upon request